QP Code **Reg.No : .....**

Name :.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION

SEMESTER V

DSC- MG5DSCZGY301-CELL AND MOLECULAR BIOLOGY

MODEL QUESTION PAPER

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Max. Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A

Answer all questions. Each question carries 1 mark each

1.Function of t-RNA during translation is [Remember]/[CO1]

A) Form the structural core of the ribosome

B) Catalyse formation of peptide bonds

C) Carry specific amino acid to specific anticodon of mRNA

D) Act as template for DNA synthesis

2. In Danielli -Davson Model of plasma membrane the lipid layer is placed between [Understand]/[CO2]

A) Two carbohydrates layers B) Two protein layers

C) Two nucleic acid layers D) One carbohydrate and one protein layer

3.The capacity of cancer cells to break away from primary growth and carried by blood or lymph to other parts of the body is [Understand]/[CO2]

A) Monoclonal origin B) Benign C) Metastasis D) Fibronectin

4. Cytokines are a type of [Remember]/[CO1]

A) Paracrine signalling molecules B) Autocrine signalling molecules

C) Endocrine signalling molecules D) None of the above

5. In molecular term muton is [Understand]/[CO2]

A) Single nucleotide pair B) Whole gene

C) Set of 3 codons D) Chromosome segment

6. Cell theory was proposed by [Remember]/[CO1]

A) Daniel and Davson C) Robert Hook and Hertwig
B) J.D.Watson and F.H.C Crick D) M.J. Schleiden and T.Schwann

7.In ABO blood grouping system, cell recognition is determined by [Understand]/[CO2]

A) Lipid bilayer on cell surface B) Carbohydrate on glycocalyx

C) Intracellular hormone receptors D) Cytoskeletal elements

8.The first two bases of a codon have a precise pairing with the bases of the anti-codon of t-RNA, while pairing of between the third base of the codon may base pair with more than one type of codon, this concept is called [Understand]/[CO2]

A) Degeneracy B) Universality C) Unambiguous D) Wobble

9. Prions are made of [Understand]/[CO2]

A) A small piece of DNA wrapped in protein
B) A single strand of RNA without a protein coat
C) A misfolded protein without any nucleic acid (DNA or RNA)
D) A lipid membrane encloses host cell enzymes

10.During osmosis, water tend to move across a plasma membrane towards [Understand]/[CO2]

A) Higher solute concentration B) Lower solute concentration

C) Equal solute concentration D) All are correct

11. Ribosome production factory of the cell is [Understand]/[CO2]

A) Nucleolus C) Nucleus
B) Chromatin D) Endoplasmic reticulum

12.During translation, the ribosome moves along the m RNA in the direction of [Understand]/[CO2]

A) 3’ → 5’ B) 5’ → 3’ C) 5’ → 5’ D) 3’ → 3’

[1x12 = 12]

Part B

Short Answer Questions. Answer any **6** questions. Each question carries **3** marks

13. Describe the structural organization of nuclear pore complex. [Understand]/[CO2]

14.Give an account on the facilitated diffusion [Understand]/[CO2]

15.State the characteristic features of genetic code [Remember]/[CO1]

16.Comment on helicase, ligase and Okazaki fragments [Understand]/[CO2]

17.Contrast desmosomes and gap junctions [Remember]/[CO1]

18. List out the functions of Golgi complex. [Remember]/[CO1]

19.Illustrate reverse transcription with an example [Understand]/[CO2]

20. Explain the basic structural frame work of microtubule. [Understand]/[CO2]

[3x6 = 18]

Part C

Short Essay Questions. Answer any **4** questions. Each question carries **5** marks

21.Compare active transport, passive transport and bulk transport [Understand]/[CO2]

22. Describe DNA replication in prokaryotes [Understand]/[CO2]

23.Summarise the electron transport chain of mitochondria [Understand]/[CO2]

24. Define Operon concept. Explain inducible operon with suitable example [Understand]/[CO2]

25. Illustrate the cell signalling pathway by cAMP with suitable example [Understand]/[CO2]

26.Discuss in detail the events of cell cycle. [Understand]/[CO2]

[5x4 = 20]